



JAN KÜSTERM.SC.

RESEARCH SOFTWARE ENGINEER

*Bremen, Germany

I am a Digital Media graduate with 10+ years of experience as research software engineer. My main focus is on production-grade web applications, involving real world users for long-term evaluations in the field. For this I leverage primarily web standards, as well as available modern web-technologies.

PUBLICATIONS

Belli, A., Küster, J., Matayeva, L., Hohmann, F., Sinner, P., Krüger, G., Wolf, K., & Hepp, A. (2025). OpenQDA (1.0.0). Zenodo.

Küster, J. (2024, March 6). Software Applications for Individuals with Low Literacy - Key Insights and Takeaways in Research Software Engineering from an Interdisciplinary Project. 4th conference for Research Software Engineering in Germany (deRSE24), Julius-Maximilians-Universität Würzburg. Zenodo.

Küster, J., Meyer, I. A. M., Windler, M., Karsten, D. Wolf, & Koppel, I. (2023). leaonline/leaonline-app: 1.1.0 (v1.1.0). Zenodo.

RELEVANT REPOSITORIES

OpenQDA -
lea.Online -
Node OAuth -

RESEARCH PROJECTS

OpenQDA - A Free Sustainable Qualitative Research Software

ZeMKI, University of Bremen | 2023 - Today

Goal: Develop a free and alternative to commercial QDA software

Key responsibilities: System architecture, software design, (web-) client architecture and dependencies management.

Major activity: Frontend implementation of rich interactive UI for qualitative data analysis using Vue3, Vite and TailwindCss.

Secondary activities: Managing Github repositories, documentation, CI, code quality assurance.

Available via

IMPACT

University of Bremen | 2023 - 2025

Goal: Implement AI-based services as key tools in formative and summative assessment activities at the University of Bremen

Key responsibilities: System architecture, software design, infrastructure, legal implications, GDPR, ethical considerations, requirements engineering

Major activities: Requirements engineering and consulting

Secondary activities: AI tool development using Python and respective libraries

lea.online

University of Bremen | 2019 - 2023

Goals: Develop a multi-application system to allow diagnostics and learning of adults with low literacy

Key responsibilities: System architecture, software design, infrastructure, legal implications, GDPR, ethical considerations, outsourcing development, requirements engineering

Major activities: web-app development, mobile app development (react native + expo)

Secondary activities: GitHub repository management and maintenance

Publications:

Meyer, I. A. M., Wolf, K. D., Windler, M., & Küster, J. (2024). Digitale berufsfeldbezogene Förderung von Literalität und Numeralität in der arbeitsorientierten Grundbildung mit der lea.App. In K.-H. Gerholz, S. Annen, R. Braches-Chyrek, J. Hufnagl, & A. Wagner (Eds.), bwp@ Spezial HT2023 Fachkräftesicherung – Zukunftsweisende Qualifizierung, gesellschaftliche Teilhabe und Integration durch berufliche Bildung.

CARO - Care Reflexion Online

University of Bremen | 2016 - 2019

Goals: Develop and evaluate a interactionistic classroom management system for German nursing education using .

Key responsibilities: System architecture, software Design, (web-) client architecture and dependencies management.

Major activity: Fullstack implementation of entire software

Secondary activities: Managing Github repositories, documentation, CI, code quality assurance.

Publications:

Darmann-Finck, I., Wolf, K. D., Schepers, C., & Küster, J. (2021). Digital unterstütztes Lernen in der Pflegeausbildung: Die Care Reflection Online (CARO) – Lernumgebung. MedienPädagogik: Zeitschrift Für Theorie Und Praxis Der Medienbildung, 317–345.

Schepers, C., & Küster, J. (2020). Sind Pflegeschulen bereit für den Einsatz digitaler Lehr-Lernformate? Berufsbildung, 181.

Project blog: (German)

DomPL-IK

University of Bremen, part time | 2012 - 2014

Invented a flexible assessment framework, targeting industrial trainees

Supervised software development lifecycle, Recruited team members

Publications:

Seifried, J., Rausch, A., Kögler, K., Brandt, S., Eigenmann, R., Schley, T., Siegfried, C., Egloffstein, M., Küster, J., Wuttke, E., Sembill, D., Martens, T., Wolf, K. D. (2016). Problemlösekompetenz angehender Industriekaufleute – Konzeption des Messinstruments und ausgewählte empirische Befunde (DomPL-IK). In K. Beck/M. Landenberger/F. Oser (Hrsg.): Technologiebasierte Kompetenzmessung in der beruflichen Bildung – Ergebnisse aus der BMBF-Förderinitiative ASCOT. Bielefeld:WBV, 119-138

Project Management Simulation Training

Getoq Consulting | 2011 / 11

Performed a two-day project simulation from management perspective

Topics included customer contracts, change management, controlling, operational tasks

Student Assistant / Programmer

University of Bremen | 2010 - 2011

Realized an online diagnosis platform for workforce literacy development (Flex)

Modeled software design, implemented various prototypes, conducted usability tests

Publications:

Koppel, I., Küster, J., & Wolf, K. D. (2013). Usability testing with female functional illiterates, Usability-testing mit funktionalen analphabetinnen. Lecture Notes in Informatics (LNI), DEFLI Proceedings - Series of the Gesellschaft Für Informatik (GI), P-218, 293–296.

EDUCATION

Graduated as M.Sc. Digital Media

University of Bremen | 2015 / 07

Master Thesis: Semi Automated Scoring in Technology Based Assessment

Developed and evaluated an algorithm for semi automated scoring of spreadsheet data

Master Project - PrIMA

University of Bremen | 2012 - 2013

Co-Invented a touch table application for medical support, co-developed software (Java)

Formed a scrum team, maintened project dev server (Debian), surveyed target audience

Master Studies Digital Media

University of Bremen | 2012 - 2015

Inter-cultural classes in English, covering special topics in computer science and design

Professionalized in research methods, software development and e-assessment

Semester Abroad

University of Melbourne | 2009 - 2010

Mastered six months of study and trans-cultural experience in Melbourne, Australia

Finished machine programming, information visualization, professional essay writing

Bachelor Studies Digital Media

University of Bremen | 2007 - 2011

Foundations of Digital Media, Computer Science and Software Engineering